

Electronic supplementary material for

“Antisocial but inconsequential: large language model agents punish cooperators yet neither sustain cooperation nor reproduce its cross-cultural geography”

This supplement reports (i) the computational and reproducibility details, (ii) the full statistical tables underlying the main text, including effect sizes, confidence intervals, the complete deviation-binned punishment, the per-society breakdown and the dispositional analysis, and (iii) the verbatim English prompt templates for both stages. All numbers are produced by `analysis/analyze.py`.

1 Reproducibility and computational details

We simulate the public-goods game with costly peer punishment of Herrmann *et al.* [1] on the FAIRGAME framework [2]. Four agents are matched as partners for ten periods. Each period every agent receives a 20-token endowment and a contribution prompt, and returns reasoning terminated by `CONTRIBUTION = X` ($0 \leq X \leq 20$); in treatment P a second prompt elicits a punishment vector `DEDUCT: A=a B=b C=c` ($0 \leq a, b, c \leq 10$). Co-players are re-labelled every period to block identity tracking, and decisions within a stage are gathered in one lockstep batch.

Each model is served with vLLM at temperature 0.8 using a fixed random seed (20080307); all runs were executed offline on a single NVIDIA RTX PRO 6000 (96 GB). The design crosses two treatments (N, P) with a language block (five languages, neutral disposition), a disposition block (four dispositions, English) and a society block (sixteen neutral city personas, English), ten independent groups per cell, sharing the English–neutral–no-society cell; this yields 480 games per model and 1440 in total. Punishment is classified by the contribution deviation between target and punisher: deviation ≥ 0 (the target gave at least as much as the punisher) is antisocial, deviation < 0 is punishment of free-riding. The format non-compliance rate is logged for every cell.

Table S1: Models evaluated. All are open-weight instruction-tuned checkpoints served identically via vLLM at temperature 0.8.

Short name	Hugging Face identifier	Reference
Qwen2.5-7B	Qwen/Qwen2.5-7B-Instruct	[3]
Gemma-2-9B	google/gemma-2-9b-it	[4]
Llama-3.1-8B	meta-llama/Llama-3.1-8B-Instruct	[5]

2 Full statistical tables

Table S2 reports the punishment effect on cooperation (P–N) with Hedges g and rank-biserial r effect sizes, Holm-adjusted Mann–Whitney p -values, 95% confidence intervals, TOST Welch equivalence tests [6] with bounds ± 2.14 tokens (half the human grand-mean gain) and JZS Bayes factors BF_{01} [7]: equivalence to a sub-human-scale effect is established for Qwen and Llama (TOST $p \leq 0.010$; $\text{BF}_{01} = 2.3$ and 2.5), while only Gemma differs from zero—in the negative direction ($g = -2.93$, $r = -0.94$, $\text{BF}_{01} = 1.8 \times 10^{-4}$). Table S3 gives the full deviation-binned punishment for all three models, showing that only Gemma reproduces the human monotone free-riding gradient. Table S4 reports the cross-societal correlations with Fisher- z confidence intervals and Fieller-corrected Fisher- z tests against the human coupling; none approaches the human $\rho \approx -0.90$, and every model’s

Table S2: Punishment effect on cooperation with effect sizes and evidence for the institutional null: Hedges g and rank-biserial correlation r for the P–N contrast on the ten per-group means, Holm–Bonferroni adjusted Mann–Whitney p -values, the TOST Welch equivalence test with bounds ± 2.14 tokens (half the human grand-mean gain; $p < 0.05$ establishes that the institutional effect is reliably smaller than half the human effect), and the JZS Bayes factor BF_{01} in favour of no P–N difference (values > 1 favour the null). Human grand-mean gain $\approx +4.3$ tokens [1].

Model	Δ (P–N)	Hedges g [95% CI]	r	MWU p (Holm)	TOST p	BF_{01}	Antisocial
Qwen2.5-7B	-0.36	-0.22 [-1.06, 0.62]	-0.11	0.705 (1.000)	0.010	2.29	59%
Gemma-2-9B	-3.55	-2.93 [-4.20, -1.66]	-0.94	0.000 (0.001)	0.993	0.00	55%
Llama-3.1-8B	+0.08	0.06 [-0.78, 0.90]	+0.04	0.910 (1.000)	0.002	2.50	54%

Table S3: Mean punishment expenditure by contribution deviation (target minus punisher) for the English, neutral baseline, treatment P. Deviations < 0 are punishment of free-riding; deviations ≥ 0 are antisocial.

Model	≤ -11	-10 to -1	0	1 to 10	11 to 20
Qwen2.5-7B	2.78	2.42	2.29	2.40	2.15
Gemma-2-9B	3.71	2.45	1.40	2.17	2.06
Llama-3.1-8B	2.84	2.86	2.60	2.78	2.53

coupling differs from it significantly ($p \leq 0.0011$). Table S5 gives the full per-society contributions for the primary model against the human pools, and Table S6 the dispositional breakdown, in which a *vengeful* prompt raises the antisocial share without restoring the institutional function of punishment.

3 Prompt templates (English)

The verbatim English contribution-stage and punishment-stage prompts are reproduced below; the four non-English versions are token-for-token translations preserving all placeholders, available in the project repository. Placeholders in braces are substituted at run time; {SOCIETY_BLOCK} and {PERSONALITY_BLOCK} are empty in the neutral, no-society condition.

Contribution stage

You are {PLAYER_ID}, one of {N_PLAYERS} members of a group. The same {N_PLAYERS} members stay together for the whole game.

The game lasts {N_ROUNDS} periods. At the start of EVERY period each member receives a fresh endowment of {ENDOWMENT} tokens. In each period you privately and simultaneously decide how many of your {ENDOWMENT} tokens to put into a shared group project; you keep the rest in your private account.

How tokens become income each period:

- Every token you keep in your private account is worth 1 token of income to you.
- Every token placed in the group project is shared among all {N_PLAYERS} members: each member receives {MPCR} tokens of income for EACH token in the project, no matter who put it there.

You contribute {ACTION_DESC} to the group project this period.

The other members are shown to you only under temporary labels ({OTHER_LABELS}). These labels are reassigned every period and do not tell you who is who across periods. You cannot communicate in any way, and all decisions are anonymous. {TREATMENT_BLOCK}

After each period you are shown how many tokens each member placed in the project that period, the total in the group project, and your own income for that period. {SOCIETY_BLOCK}

Table S4: Cross-societal structure (treatment P, 16 society personas) with Fisher- z 95% confidence intervals. $p_{\text{vs. hum}}$ is the two-sided Fisher- z test (Fieller-corrected) that the model’s antisocial-cooperation coupling differs from the human $\rho \approx -0.90$ over the same 16 pools [1].

Model	$\rho_{\text{anti,coop}}$ [95% CI]	$p_{\text{vs. hum}}$	$\rho_{\text{LLM,HUM}}$ [95% CI]	Wilcoxon p	$\overline{\text{LLM}} - \overline{\text{HUM}}$
Qwen2.5-7B	-0.09 [-0.56, +0.43]	0.0006	+0.33 [-0.20, +0.71]	0.669	+0.80
Gemma-2-9B	-0.15 [-0.60, +0.37]	0.0011	+0.06 [-0.45, +0.54]	0.005	-3.30
Llama-3.1-8B	+0.30 [-0.23, +0.69]	0.0000	+0.17 [-0.35, +0.62]	0.025	-2.47

Table S5: Per-society contributions for Gemma-2-9B against the human pools of Herrmann *et al.* [1]; antisocial = total antisocial punishment points (treatment P). Societies ordered by human treatment-P contribution.

Society	LLM N	LLM P	Antisocial	Human N	Human P
Boston	13.52	9.70	1487	9.3	18.0
Copenhagen	13.00	9.05	1404	11.5	17.7
St.Gallen	13.97	9.78	1460	10.1	16.7
Zurich	13.33	9.50	1496	9.3	16.2
Nottingham	13.12	9.49	1446	6.9	15.0
Seoul	13.44	9.57	1346	7.9	14.7
Bonn	12.97	9.97	1293	9.2	14.5
Melbourne	13.19	9.73	1393	4.9	14.1
Chengdu	13.41	8.94	1374	8.0	13.9
Minsk	12.34	9.70	1589	10.5	12.9
Samara	12.60	9.81	1507	9.7	11.7
Dnipropetrovsk	13.73	9.87	1470	10.6	10.9
Muscat	13.90	9.42	1472	10.0	9.9
Istanbul	13.64	9.77	1218	5.4	7.1
Riyadh	13.46	9.46	1477	7.6	6.9
Athens	13.90	9.36	1592	6.4	5.7

{PERSONALITY_BLOCK}

CURRENT SITUATION

This is period {CURRENT_ROUND} of {N_ROUNDS}.

Results of the periods played so far:

{HISTORY_BLOCK}

YOUR DECISION

Decide how many of your {ENDOWMENT} tokens to place in the group project this period ({ACTION_DESC}). You may reason briefly first. Then finish your reply with EXACTLY one line in this format, and nothing after it:

CONTRIBUTION = X

(where X is an integer from {CONTRIB_MIN} to {CONTRIB_MAX})

Punishment stage (treatment P)

You are {PLAYER_ID}. This is the second stage of period {CURRENT_ROUND} of {N_ROUNDS}.

This period you placed {MY_CONTRIB} tokens in the group project. The total placed by all members was {GROUP_PROJECT_TOTAL}, and your income from the first stage this period is {MY_INCOME_THIS_PERIOD} tokens.

Here is how much each of the other members placed in the project this period:

{OTHERS_BLOCK}

Table S6: Dispositional steerability (English, treatment P): mean contribution and antisocial share of punishment by disposition prompt and model.

Model	Disposition	Mean contribution	Antisocial share
Qwen2.5-7B	Neutral	13.90	59%
	Cooperative	16.70	64%
	Selfish	8.43	63%
	Vengeful	12.28	63%
Gemma-2-9B	Neutral	9.86	55%
	Cooperative	14.28	61%
	Selfish	6.41	59%
	Vengeful	8.39	66%
Llama-3.1-8B	Neutral	10.75	54%
	Cooperative	11.95	57%
	Selfish	8.40	56%
	Vengeful	10.70	56%

You may now assign deduction points to each of these members, simultaneously and anonymously. For EACH member you choose a whole number of deduction points from 0 to {MAX_PUNISH}.

- Each deduction point you assign costs YOU {PUNISH_COST} token.
- Each deduction point you assign reduces THAT member's income this period by {PUNISH_IMPACT} tokens.

The other members decide at the same time and may assign deduction points to anyone, including you. No one is told who assigned points to whom; you are only told the total number of points you received.

{RECEIVED_LAST_PERIOD}

{SOCIETY_BLOCK}

{PERSONALITY_BLOCK}

DEDUCTIONS SO FAR

{PUNISH_HISTORY_BLOCK}

YOUR DECISION

Decide how many deduction points (0 to {MAX_PUNISH}) to assign to each member shown above. You may reason briefly first. Then finish your reply with EXACTLY one line in this format, and nothing after it:

DEDUCT: A=a B=b C=c

(where a, b, c are integers from 0 to {MAX_PUNISH}, for Member A, Member B, Member C)

References

- [1] Benedikt Herrmann, Christian Thöni, and Simon Gächter. “Antisocial punishment across societies”. In: *Science* 319.5868 (2008), pp. 1362–1367. DOI: 10.1126/science.1153808.
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- [3] Qwen Team. *Qwen2.5 Technical Report*. arXiv:2412.15115. 2025. arXiv: 2412.15115 [cs.CL].
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